CASE STUDY | BMW GROUP

BMW Production Network & Working Capital Analysis

An independent public-data study of model-site redundancy, drivetrain breadth, battery localisation and the Neue Klasse transition

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Research completed	10 August 2026
Core evidence cut-off	10 August 2026
Baseline	BMW Group 2025 production programme
Publication date	28 August 2026

Executive findings

The study begins with BMW Group's 2025 production-programme table and reconstructs the plant-model network from the published model allocation. It then overlays transition events available by 10 August 2026 and adds the Gen6 battery layer. The earlier working-capital analysis is deliberately retained only as a downstream diagnostic. (BMW Group, 2026a; BMW Group, 2025c; BMW Group, 2025a; BMW Group, 2026h; BMW Group, 2026g; BMW Group, 2026b; BMW Group, 2026i)

14	27	63	12
comparable sites BMW/MINI production sites used in the model matrix	model lines Named model lines after excluding non-specific "BMW M" from the count	model-site links One link means a model is listed at a site	single-site lines 44.4% of the comparable model lines

What the network looks like

1	**The network is becoming hybrid, not purely specialised. BMW is adding dedicated BEV capability at Debrecen and moving Munich towards BEV-only production, while established sites such as Dingolfing and Regensburg retain mixed-drivetrain flexibility. (BMW Group, 2025c; BMW Group, 2026f; BMW Group, 2026e; BMW Group, 2026c)**
2	**Model concentration remains visible. Twelve of 27 model lines are listed at only one comparable site. That is a structural dependency flag, not a forecast of disruption or a value-at-risk estimate.**
3	**Battery localisation tightens regional coupling. BMW's Gen6 plan places five new battery assembly sites close to major vehicle-production regions: Irlbach-Straßkirchen, Debrecen, Woodruff, Shenyang and San Luis Potosí. (BMW Group, 2025a; BMW Group, 2024b; BMW Group, 2026i; BMW Group, 2024a)**
4	**H1 2026 supplies a live downstream signal. Automotive production exceeded deliveries by 110,784 vehicles in H1 2026 versus 31,014 in H1 2025; inventory increased by €2.801bn and BMW reported a €1.858bn negative change in working capital. (BMW Group, 2026h; BMW Group, 2026g)**

Core conclusion **BMW is building a two-speed production architecture: a flexible multi-drivetrain backbone remains valuable, while dedicated or BEV-focused nodes and regional battery systems are added around it. The resulting management issue is not "flexibility versus specialisation" in isolation, but how the two architectures interact during product and plant transitions.**

How to read the evidence **Source-reported facts are tagged with source IDs. Project calculations are explicitly described as calculations. Structural classifications are analytical tests created for this study. Financial figures are used as a downstream signal and are not presented as proof of causation.**

1. Research question, scope and method

1.1 Why this question

BMW describes flexibility and market proximity as deliberate features of its production network. The key analytical issue is therefore not whether BMW is electrifying, but how the network is being re-architected while established flexible plants still absorb existing programmes. The Neue Klasse introduces dedicated or BEV-focused nodes, while other sites receive new technology inside mixed-drive production systems. (BMW Group, 2025c; BMW Group, 2026f; BMW Group, 2026e)

This produces a more specific public-data question: when flexibility is redistributed rather than simply removed, where does redundancy improve, where does it thin out, and where do new battery and vehicle dependencies become visible? The study

answers those questions with plant-model counts, a model-site matrix, a transition overlay and a limited H1 2026 financial cross-check.

1.2 Analytical design

Analytical layer	What is tested	Why it is included
Starting signal	H1 2026 production > deliveries; inventory and working-capital pressure	Why the original project mattered, but not the full explanation. (BMW Group, 2026h; BMW Group, 2026g)
Structural question	Where is flexibility retained, where is model allocation concentrated, and where are new specialised nodes appearing?	The core question tested with public plant/model data.
Network layer	Plant-model links + drivetrain breadth + transition events + battery nodes	Creates a single operating-network view.
Financial cross-check	H1 2026 production gap, inventory movement, working capital and FCF	Tests whether the network story sits beside a live operating signal; no causality claimed.
Layer	**Input**	**Treatment**
Baseline	BMW 2025 series-production programme	Use the published plant/model/drivetrain table as a cross-sectional snapshot. (BMW Group, 2026a)
Core unit	Model-site link	A link is counted when a named model line appears in the 2025 production programme for a comparable site.
Concentration test	Site count per model line	One site = single-site line; two or more = multi-site line. This measures structural redundancy only.
Drivetrain breadth indicator	Drivetrain breadth by site	Count the published portfolio categories: ICE, BEV and PHEV. Five comparable sites list all three. (BMW Group, 2026a)
Transition overlay	Public releases through 10 Aug 2026	Add dated evidence of plant conversion, Neue Klasse launches and battery-node build-out.
Financial signal	H1 2026 Automotive data	Use production, deliveries, inventory, working capital and FCF as downstream operating evidence. (BMW Group, 2026h; BMW Group, 2026g)

1.3 Scope decisions

The comparable matrix excludes Goodwood because the 2025 table is a Rolls-Royce production site, and excludes Debrecen because BMW's 2025 report lists the site with BMW iX1 while the Debrecen launch materials identify the iX3 as the first Neue Klasse series-production vehicle. Rather than forcing a choice, the study treats this as a source-reconciliation issue and analyses Debrecen in the transition overlay. (BMW Group, 2026a; BMW Group, 2025b; BMW Group, 2025d)

The jointly controlled Zhangjiagang (Spotlight) site is included in the model matrix because BMW lists the MINI Cooper and MINI Aceman there. It is kept separate from BMW-owned plant counts. Non-specific "BMW M" categories are excluded from model-line counts because they do not identify a single model line. (BMW Group, 2026a)

Capacity and utilisation are not estimated. BMW does not publish a comparable current capacity figure for every site in the selected sources, and model-level production volumes are not available consistently enough to calculate revenue or production-at-risk by site. That is a deliberate boundary, not a missing calculation.

2. The 2025 production network: structural baseline

BMW's 2025 report states that the production network comprised 32 plants in 15 countries and that 2,456,561 BMW Group vehicles were produced in 2025. The full network also includes motorcycle, engine and component locations; this study narrows the operating analysis to the passenger-vehicle production sites relevant to model allocation. (BMW Group, 2026a)

Site	Country	Brand	Model lines	Drive portfolio	Transition position
Araquari	Brazil	BMW	4	ICE, PHEV	No specific Neue Klasse vehicle start identified in selected sources by 10 Aug 2026 (BMW Group, 2026a)
Chennai	India	BMW	9	ICE, BEV	No specific Neue Klasse vehicle start identified in selected sources by 10 Aug 2026 (BMW Group, 2026a)
Dingolfing	Germany	BMW	7	ICE, BEV, PHEV	Established flex plant; Neue Klasse technologies entering 7 Series from Jul 2026 (BMW Group, 2026e)
Leipzig	Germany	BMW/MINI	3	ICE, BEV, PHEV	BMW/MINI flex site; ICE, BEV and PHEV listed (BMW Group, 2026a)
Munich	Germany	BMW	3	ICE, BEV, PHEV	2026 i3 launch; BEV-only configuration from 2027 (BMW Group, 2026f)
Oxford	UK	MINI	2	ICE	No specific Neue Klasse vehicle start identified in selected sources by 10 Aug 2026 (BMW Group, 2026a)
Rayong	Thailand	BMW/MINI	10	ICE, BEV, PHEV	Regional BMW/MINI hub; three drivetrain categories (BMW Group, 2026a)
Regensburg	Germany	BMW	4	ICE, BEV, PHEV	High-volume flex site; three drivetrain categories (BMW Group, 2026c)
Rosslyn	South Africa	BMW	1	ICE, PHEV	No specific Neue Klasse vehicle start identified in selected sources by 10 Aug 2026 (BMW Group, 2026a)

Site	Country	Brand	Model lines	Drive portfolio	Transition position
San Luis Potosí	Mexico	BMW	2	ICE, PHEV	Neue Klasse production planned from 2027 (BMW Group, 2024a)
Shenyang (Dadong)	China	BMW	4	ICE, BEV	China BEV/ICE site; Neue Klasse China overlay (BMW Group, 2026a; BMW Group, 2025d)
Shenyang (Tiexi)	China	BMW	6	ICE, BEV	China BEV/ICE site; plant allocation not fully resolved (BMW Group, 2026a; BMW Group, 2025d)
Spartanburg	USA	BMW	6	ICE, PHEV	X-model hub; BEV addition planned in 2026 (BMW Group, 2025c; BMW Group, 2026b)
Zhangjiagang (Spotlight)	China	MINI	2	BEV	Jointly controlled; BEV MINI production (BMW Group, 2026a)
Reading the table The table is a baseline, not a live production-control map. A model appearing at a site means BMW's 2025 report listed that model there; it does not disclose how much of the model was produced there or whether the allocation has since changed.					

2.1 Where the network is broadest

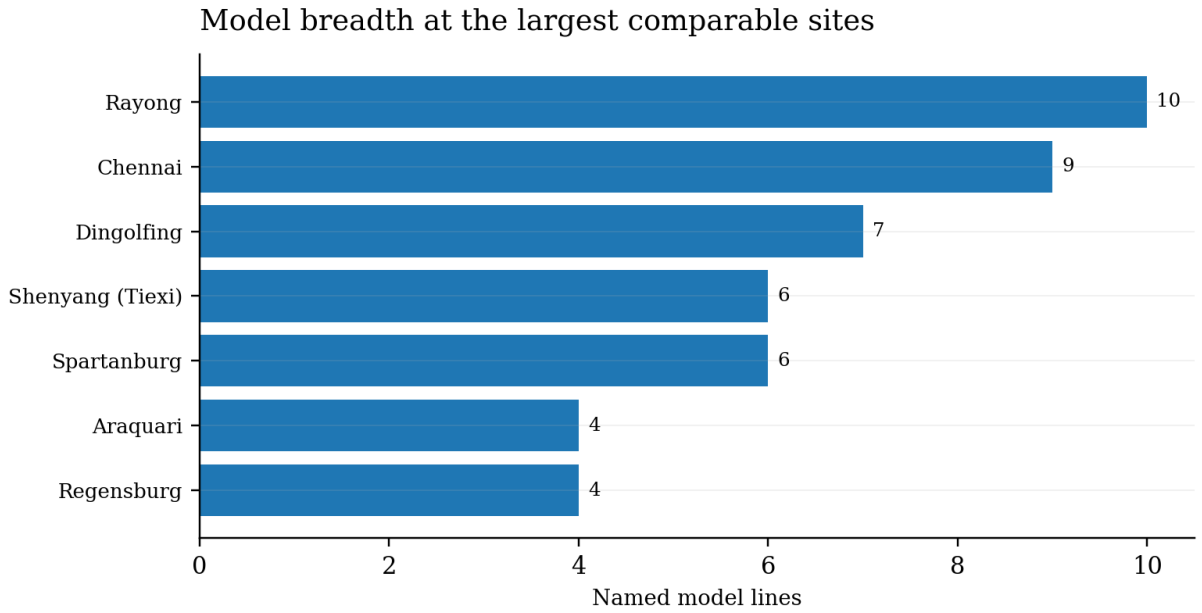


Figure 1. Named model breadth at the largest comparable sites, based on the 2025 BMW Group production programme. (BMW Group, 2026a)

Rayong and Chennai have the broadest named model portfolios in the comparable matrix. Dingolfing is narrower in model count but strategically important because its portfolio combines luxury models with BEV and conventional drivetrains, making the plant a useful example of BMW's continuing flexibility model. (BMW Group, 2026a; BMW Group, 2026e)

3. Model concentration: where is redundancy thin?

The most tangible original test in the study is simple: how many comparable sites are listed for each named model line? This is not a probability model. It is a structural redundancy measure that can be reproduced directly from the published plant table.

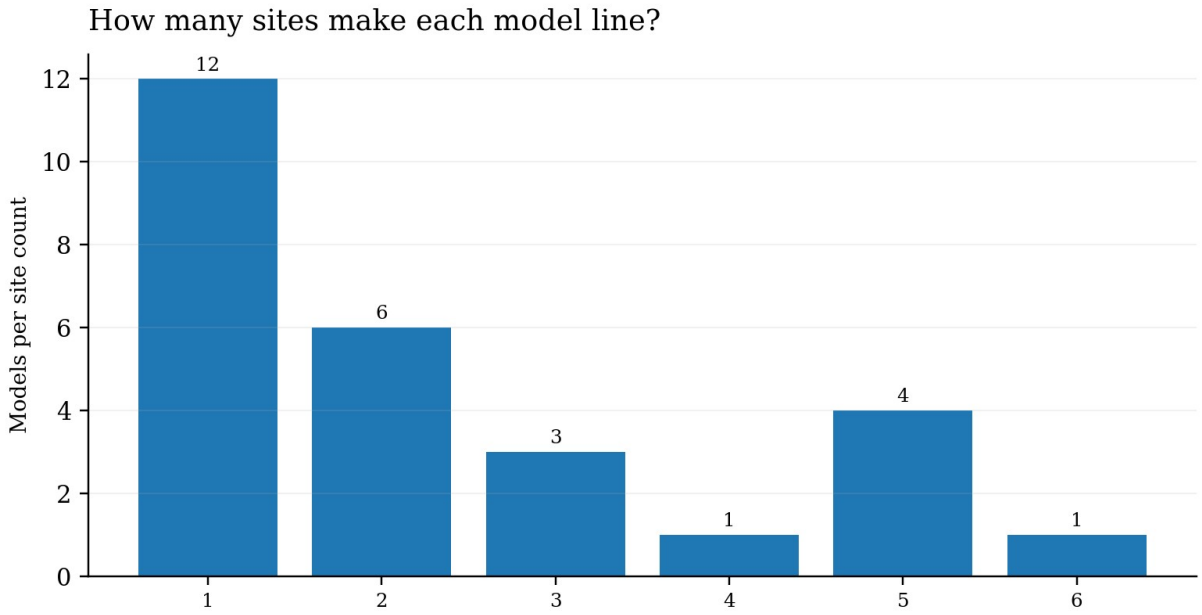


Figure 2. Distribution of comparable model lines by number of listed production sites. 12 of 27 model lines are single-site lines. (BMW Group, 2026a)

3.1 Concentration results

Listed sites per model	Model lines	Share
1 site	12	44.4%
2 sites	6	22.2%
3 sites	3	11.1%
4 sites	1	3.7%
5 sites	4	14.8%
6 sites	1	3.7%
Total	27	100%

Across the 27 model lines, there are 63 model-site links, giving a simple average of 2.33 listed sites per model line. The distribution is more informative than the average: almost half of the model lines are listed at one site, while a small group of high-volume names such as the BMW 3 Series is spread across six sites. (BMW Group, 2026a)

3.2 Which plants carry single-site lines?

Site	Single-site lines	Lines
Dingolfing	3	BMW 8 Series, BMW i7, BMW iX
Regensburg	2	BMW X2, BMW iX2
Leipzig	1	BMW 1 Series
Munich	1	BMW i4

Site	Single-site lines	Lines
Oxford	1	MINI Cooper Convertible
Shenyang (Dadong)	1	BMW iX3
Shenyang (Tiexi)	1	BMW i3
Spartanburg	1	BMW XM
Zhangjiagang (Spotlight)	1	MINI Aceman
Important limitation A single-site model line can be intentional. The public data does not provide a model-level volume, contribution margin or replacement-capacity figure that would allow these lines to be converted into economic value at risk. The correct interpretation is dependency, not predicted disruption.		

3.3 Why the concentration finding matters during the transition

The concentration pattern becomes more interesting when it is placed next to the transition architecture. Dedicated BEV sites are entering the network at the same time that some established plants continue to carry broad, mixed-drivetrain portfolios. This means redundancy can move in two directions at once: some individual models remain highly concentrated, while the overall network retains flexibility through other plants and regional hubs. (BMW Group, 2025c; BMW Group, 2026f; BMW Group, 2026e)

4. The plant-model network: the underlying structure

The matrix below is the underlying structure behind the concentration analysis. A filled cell means the model line is listed in BMW’s 2025 series-production programme for that site. Debrecen is intentionally absent because of the source discrepancy described earlier. (BMW Group, 2026a; BMW Group, 2025b; BMW Group, 2025d)

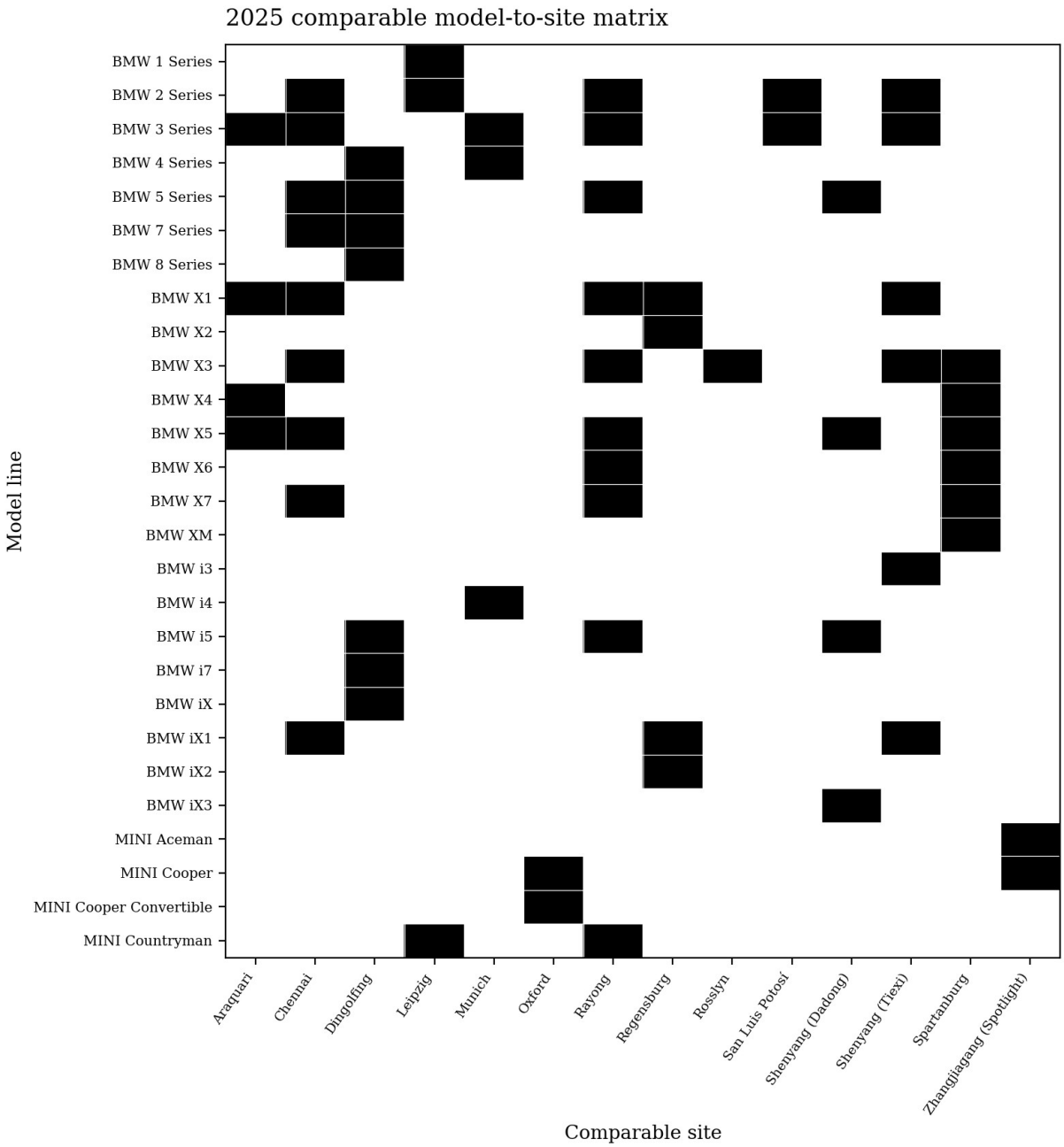


Figure 3. Comparable 2025 model-to-site matrix. The matrix is a project visualisation of BMW's published plant programme; it is not a BMW chart. (BMW Group, 2026a)

4.1 Selected high-redundancy examples

Model line	Sites	Listed locations	Reading
BMW 3 Series	6	Araquari, Chennai, Munich, Rayong, San Luis Potosí, Shenyang (Tiexi)	Highest listed site count in the comparable matrix.
BMW 2 Series	5	Chennai, Leipzig, Rayong, San Luis Potosí, Shenyang (Tiexi)	Broad regional spread across Europe, Asia and the Americas.
BMW X1	5	Araquari, Chennai, Rayong, Regensburg, Shenyang (Tiexi)	Distributed across multiple regional and flexible sites.

Model line	Sites	Listed locations	Reading
BMW X5	5	Araquari, Chennai, Rayong, Shenyang (Dadong), Spartanburg	Links the X-model portfolio to several regional production hubs.
BMW X3	5	Chennai, Rayong, Rosslyn, Shenyang (Tiexi), Spartanburg	Strong regional diversification in the baseline snapshot.

The purpose of these examples is not to rank BMW models by importance. It is to show how the network contains different redundancy patterns, which become relevant when plant roles change. A widely distributed legacy model and a single-site BEV model should not be treated as the same kind of operational dependency.

5. Where flexibility still sits in the network

BMW's own network description makes flexibility a central design principle: German plants Munich, Dingolfing, Regensburg and Leipzig each had at least one BEV in addition to conventional drives, while BMW also used flexible structures in other regions. (BMW Group, 2025c) The 2025 production table lets us quantify a public-data proxy for that idea.

Plant	Drive breadth	Model lines	Why it matters
Dingolfing	ICE + BEV + PHEV	7	5 and iX programmes plus luxury portfolio; new 7 Series adds Neue Klasse technology to an established line (BMW Group, 2026e)
Leipzig	ICE + BEV + PHEV	3	BMW and MINI models with all three drive categories listed (BMW Group, 2026a)
Munich	ICE + BEV + PHEV	3	Mixed portfolio in 2025; i3 series launch in Aug 2026, BEV-only from 2027 (BMW Group, 2026a; BMW Group, 2026f)
Rayong	ICE + BEV + PHEV	10	Largest model breadth in comparable matrix; regional hub (BMW Group, 2026a)
Regensburg	ICE + BEV + PHEV	4	High-volume X1/X2 plant; flexible drivetrain production (BMW Group, 2026a; BMW Group, 2026c)

Five of the 14 comparable sites list ICE, BEV and PHEV in the 2025 report: Dingolfing, Leipzig, Munich, Rayong and Regensburg. This is used as a public-data proxy for visible multi-drivetrain capability; it does not measure switching speed, capacity elasticity or conversion cost. (BMW Group, 2026a)

5.1 Dingolfing as a transition pattern

Dingolfing is especially useful because the 1 July 2026 7 Series launch shows Neue Klasse technologies entering an established product and production system rather than waiting for a dedicated new plant. BMW states that the new 7 Series is built on a single line with different drive variants, and the production launch incorporates Neue Klasse technologies into the established model. (BMW Group, 2026e)

5.2 Regensburg as a high-volume flexible node

BMW reported 356,901 vehicles produced at Regensburg in 2025, making it BMW Group's highest-volume car plant in Europe. The plant produces the X1, iX1, X2 and iX2 and BMW describes production across three drivetrain variants. (BMW Group, 2026c) The combination of scale, breadth and flexibility makes Regensburg a different type of network node from a dedicated BEV plant such as Debrecen.

Analytical implication BMW does not appear to be abandoning its flexible manufacturing backbone. Instead, the evidence points to a layered architecture: dedicated BEV nodes are added while flexible sites continue to absorb mixed demand and, increasingly, new technologies.

6. Neue Klasse transition: from product launch to network architecture

The transition is not a single plant move. Public BMW material shows a sequence of specialised launch nodes, live-plant conversions and regional roll-outs. The dates below provide the chronology used in the analysis.

Date	Scope	Evidence	Analytical reading
24 Mar 2025	Global network	BMW describes flexible plants as remaining important as electrification advances at different speeds.	Baseline strategy (BMW Group, 2025c)
29 Sep 2025	Debrecen	Plant officially opens; BMW identifies the iX3 as the first Neue Klasse series-production vehicle and notes on-site Gen6 battery assembly.	Dedicated BEV node (BMW Group, 2025b)
Late Oct 2025	Debrecen	BMW states that iX3 series production began in late October 2025.	Neue Klasse production launch (BMW Group, 2025b; BMW Group, 2025d)
4 Feb 2026	Munich	Pre-series BMW i3 vehicles enter production during the final preparation phase after the plant conversion.	Live-plant conversion (BMW Group, 2026d)
1 Jul 2026	Dingolfing	New BMW 7 Series enters series production with Neue Klasse technologies integrated into an established multi-drivetrain plant.	Technology diffusion (BMW Group, 2026e)
6 Aug 2026	Munich	BMW i3 enters series production; from 2027 Plant Munich will exclusively produce fully electric vehicles.	BEV-only transition path (BMW Group, 2026f)
2026	Shenyang	China-specific Neue Klasse iX3 rollout is part of the regional production programme; public materials do not resolve every plant-level allocation.	Regional localisation (BMW Group, 2025d)
Before end-2026	Spartanburg / Woodruff	BMW planned U.S. iX5 assembly before the end of 2026; Woodruff provides local high-voltage battery assembly capability.	Flex hub adds battery coupling (BMW Group, 2026b; BMW Group, 2026i)
2027	San Luis Potosí	Neue Klasse production planned from 2027, supported by local high-voltage battery assembly.	Future regional BEV node (BMW Group, 2024a)
2027	Munich	BMW states Plant Munich will exclusively produce fully electric vehicles from 2027.	Dedicated BEV operating model (BMW Group, 2026f)

6.1 The architecture that emerges

FLEXIBLE BACKBONE	**Dingolfing · Leipzig · Munich (before cut-over) · Regensburg · Rayong**	**Mixed-drivetrain production; absorbs multiple programmes and supports demand balancing.**
DEDICATED / BEV-FOCUSED NODES	Debrecen · Munich (from 2027) · Spotlight MINI	Higher specialisation; clearer BEV identity and new product/plant architectures.

FLEXIBLE BACKBONE	**Dingolfing · Leipzig · Munich (before cut-over) · Regensburg · Rayong**	**Mixed-drivetrain production; absorbs multiple programmes and supports demand balancing.**
REGIONAL BEV + BATTERY NODES	Shenyang · Spartanburg/Woodruff · San Luis Potosí	Regional production coupled with local or near-site Gen6 battery capacity.
Main finding The public evidence supports a hybrid architecture rather than a simple shift from flexible plants to dedicated EV factories. BMW is doing both: preserving flexibility where it has operating value and adding specialised nodes where the new architecture makes that sensible.		

7. Battery localisation: a second network sits underneath the vehicle network

At the 10 August 2026 evidence cut-off, BMW's Gen6 battery strategy identified five regional assembly locations: Irlbach-Straßkirchen, Debrecen, Shenyang, San Luis Potosí and Woodruff. BMW links this to its "local for local" principle. (BMW Group, 2025a; BMW Group, 2024b)

Battery location	Region	Vehicle-network relationship	Evidence
Irlbach-Straßkirchen	Germany	Near German vehicle network	Gen6 battery series production planned in H2 2026; intended to supply German plants. (BMW Group, 2025a; BMW Group, 2026f)
Debrecen	Hungary	On-site	Gen6 battery assembly integrated with the first Neue Klasse production site. (BMW Group, 2025b)
Shenyang	China	Local	Gen6 battery manufacturing supports China's Neue Klasse production. (BMW Group, 2025a; BMW Group, 2025d)
Woodruff	USA	Near Spartanburg	High-voltage battery assembly for U.S. BEV production; series production planned in Dec 2026. (BMW Group, 2026i)
San Luis Potosí	Mexico	On-site / planned	High-voltage battery assembly being integrated with planned 2027 Neue Klasse vehicle production. (BMW Group, 2024b; BMW Group, 2024a)

7.1 Why this matters for network analysis

The battery layer changes the interpretation of plant flexibility. A vehicle plant can be operationally flexible while still depending on a more specialised battery supply node. Conversely, local battery production can reduce some forms of transport dependency while creating stronger coupling between the timing of battery ramp-up and vehicle ramp-up. The public data supports the existence of these links, but not a quantitative disruption model.

What cannot be calculated The sources do not disclose a comparable battery capacity, supplier allocation, inventory buffer or plant-level outage probability for all five regions. The study therefore maps the architecture but does not assign a percentage reduction in supply risk.

8. Three plants, three transition patterns

8.1 Munich: conversion under live production

Munich is the clearest example of a network transition that changes the identity of an existing plant. BMW says the site was transformed over four years while production continued, with a new body shop, assembly hall and logistics areas. On 6 August 2026, the new BMW i3 entered series production and BMW stated that from 2027 the plant will exclusively produce fully

electric vehicles. BMW also reported a further 10% reduction in manufacturing costs at the start of i3 production. (BMW Group, 2026d; BMW Group, 2026f)

Observable fact	Why it matters
Four-year transformation while production continued (BMW Group, 2026f)	Change is being absorbed inside an operating asset rather than through a greenfield-only strategy.
BEV-only production from 2027 (BMW Group, 2026f)	The plant moves from mixed drivetrain flexibility towards specialisation.
Battery from Bavaria; e-drive from Steyr (BMW Group, 2026f)	Neue Klasse creates explicit regional production linkages between sites.

8.2 Dingolfing: Neue Klasse technology without a BEV-only plant

Dingolfing shows the opposite design choice. BMW's July 2026 7 Series launch integrates Neue Klasse technologies into an established production line. BMW says all drivetrain variants are built flexibly on the same line with the 5 Series and iX. (BMW Group, 2026e) This is important because it demonstrates that "Neue Klasse" is not synonymous with "dedicated EV factory".

8.3 Regensburg: scale plus flexibility

Regensburg produced 356,901 vehicles in 2025, the highest volume of any BMW Group car plant in Europe. BMW reports that more than 150,000 of those vehicles were electrified and that X1/X2 production uses flexible drivetrain variants. (BMW Group, 2026c) The site therefore acts as a useful counterpoint to Debrecen: high volume, broad product portfolio and mixed technology rather than dedicated BEV specialisation.

Case-study reading These sites are not presented as "good" or "bad". They represent three different industrial roles: conversion to specialisation (Munich), technology diffusion within flexibility (Dingolfing), and high-volume mixed production (Regensburg).

9. H1 2026: the downstream operating signal

The original working-capital analysis is retained here because it provides a downstream test of the network story. BMW reported that H1 2026 Automotive production exceeded deliveries by 110,784 vehicles, compared with 31,014 in H1 2025. This is a project calculation from BMW-reported production and delivery totals. (BMW Group, 2026h)

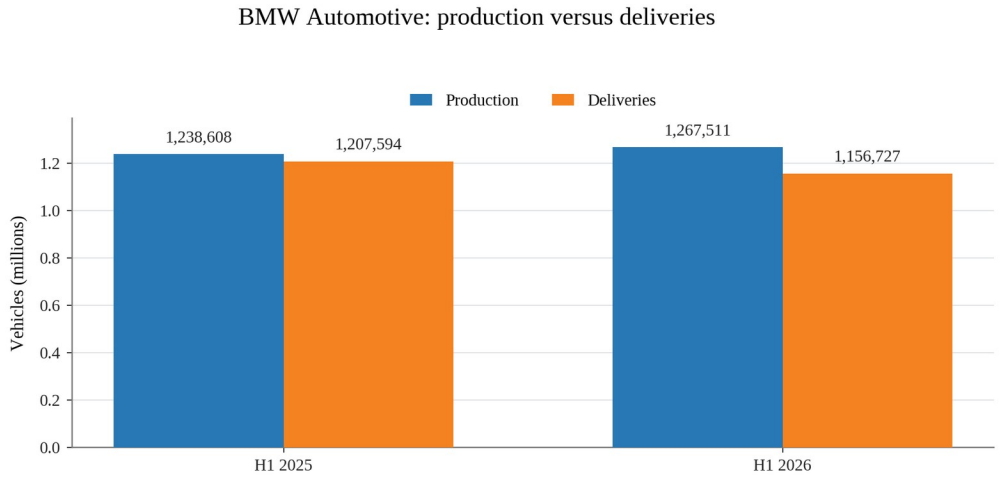


Figure 4. Automotive production and deliveries, H1 2025 versus H1 2026. Source: BMW H1 2026 report; production-minus-deliveries is calculated in this project. (BMW Group, 2026h)

Metric	Current period*	Comparison period*	Change
Production	1,267,511	1,238,608	+28,903
Deliveries	1,156,727	1,207,594	−50,867
Production − deliveries	110,784	31,014	+79,770
Automotive inventory, period end	€22.171bn	€19.370bn*	+€2.801bn*
Change in working capital	−€1.858bn	−€0.252bn	−€1.606bn
Automotive free cash flow	€1.290bn	€2.345bn	−€1.055bn

* Period basis: production, deliveries, working capital and free cash flow compare H1 2026 with H1 2025. Automotive inventory compares 30 June 2026 with 31 December 2025. Working capital and free cash flow are BMW-reported values. (BMW Group, 2026h)

9.1 Complete free-cash-flow reconciliation

FCF bridge	H1 2026	H1 2025
Operating cash inflow	€4.786bn	€7.292bn
Investing cash flow	−€3.310bn	−€4.557bn
Adjustment for net investment in marketable securities and investment funds	−€0.186bn	−€0.390bn
Automotive free cash flow	€1.290bn	€2.345bn

BMW’s management commentary says the Q2 working-capital effect reduced Automotive free cash flow by around €1.3bn because inventories built up as production exceeded sales, and that production planning had been adjusted to bring inventories back towards the prior-year level by year-end. (BMW Group, 2026g) This is the point where the old working-capital study feeds into the new network study: it is evidence of an operating mismatch, not proof of a network-design cause.

How this connects to the main study This financial signal does not prove that the Neue Klasse transition caused the inventory build. It shows that, during the same public transition period, BMW experienced a larger production-versus-deliveries gap and a material working-capital drag. The network analysis treats that as a downstream coordination signal, not as causal evidence.

10. What the combined evidence says

Finding 1 — BMW is building a hybrid network

The plant-level evidence points to a hybrid architecture. Debrecen and the future Munich configuration are increasingly specialised for BEV production, while Dingolfing, Leipzig, Regensburg and Rayong retain broader mixed-drivetrain portfolios. The network therefore does not move from one operating model to another; it adds specialised nodes around a flexible backbone. (BMW Group, 2026a; BMW Group, 2025c; BMW Group, 2026f; BMW Group, 2026e)

Finding 2 — redundancy is uneven, not absent

The 2025 baseline contains 12 single-site model lines and 15 multi-site lines. The important result is the dispersion of dependency: some models are spread across several sites, while others rely on one listed location. Seven of the 12 single-site lines are BEV names in the published 2025 table, but model naming alone does not establish that every one is a Neue Klasse generation; the iX3 is an explicit example of why the transition overlay is necessary. (BMW Group, 2026a)

Finding 3 — battery localisation creates a second dependency map

The vehicle-site matrix sits on top of a second network: BMW’s Gen6 battery assembly footprint. Five planned/operating locations - Irlbach-Straßkirchen, Debrecen, Shenyang, San Luis Potosí and Woodruff - connect regional battery production with the vehicle network. This can shorten physical supply distances, but it also couples battery ramp-up timing with vehicle ramp-up timing. Public evidence supports the linkage; it does not support a quantitative disruption probability. (BMW Group, 2025a; BMW Group, 2024b; BMW Group, 2026i; BMW Group, 2024a)

Finding 4 — transition risk is partly about coordination, not only technology

The central coordination issue is the coexistence of two operating logics: flexible plants that can absorb different drivetrain programmes, and specialised nodes designed around the Neue Klasse. H1 2026 shows why that matters operationally: production exceeded deliveries by 110,784 vehicles and BMW reported a €1.858bn negative change in working capital. Those observations are associated with the transition period, but the project does not claim that Neue Klasse caused them. (BMW Group, 2026f; BMW Group, 2026e; BMW Group, 2026h; BMW Group, 2026g)

11. What to monitor as the transition develops

Monitor	Question	Why it matters
Model redundancy	Do newly launched Neue Klasse models remain single-site for long, or are second and third production locations added?	Lower site count means higher structural dependency.
Plant role changes	Which flexible plants lose drivetrain breadth as more dedicated BEV production is introduced?	Signals whether flexibility is genuinely being reduced or merely repositioned.
Battery synchronisation	Do local/near-site battery plants ramp on the same timetable as vehicle programmes?	Misalignment could become a practical bottleneck even when vehicle assembly capacity exists.
Production-demand balance	Does the production-minus-deliveries gap narrow after H1 2026?	Provides a downstream read on inventory and working-capital pressure, without assuming a cause.
Public evidence quality	Do future disclosures resolve plant-level model allocation in Shenyang and the Debrecen iX1/iX3 discrepancy?	Better evidence would allow a more granular transition matrix.

12. Limitations and data-integrity notes

Issue	What the evidence shows	Treatment in this study
Debrecen programme discrepancy	BMW’s 2025 report lists iX1 at Debrecen; BMW’s Debrecen launch material identifies iX3 as the first Neue Klasse series-production vehicle.	Excluded from comparable model matrix; analysed in transition overlay. (BMW Group, 2026a; BMW Group, 2025b; BMW Group, 2025d)
i3 / iX3 generation naming	The 2025 table can contain model names that do not represent the new Neue Klasse generation.	The 2025 matrix is treated as a snapshot; 2026 launches are overlaid separately.
Shenyang plant-level allocation	BMW refers to the Shenyang production complex and regional Neue Klasse activity, but public material does not always assign every model to Dadong or Tiexi.	Avoided plant-level claims where evidence is not explicit. (BMW Group, 2026a; BMW Group, 2025d)
Capacity and utilisation	Comparable current capacity by plant is not published in the selected source set.	No production-at-risk, utilisation or probability-of-disruption calculation.
Model-site volume	The 2025 plant table shows programme membership, not plant-level model volume.	Site counts are structural indicators, not value-at-risk measures.
Causality	H1 2026 inventory and working-capital changes overlap with the Neue Klasse transition but are not proven to be caused by it.	Financial data is presented as a downstream signal only. (BMW Group, 2026h; BMW Group, 2026g)

12.1 What this project does not claim

- It is not an internal BMW study and contains no confidential or proprietary BMW information.
- It does not estimate lost revenue or production from a hypothetical plant outage.
- It does not predict which plant BMW will close, expand or repurpose beyond what BMW has publicly announced.
- It does not infer plant capacity, inventory buffers or supplier allocation where BMW has not disclosed them.
- It does not claim that the H1 2026 cash-flow effect was caused by Neue Klasse.

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Appendix A — Model-site concentration table

Model line	Sites	Listed sites	Single-site
BMW 3 Series	6	Araquari, Chennai, Munich, Rayong, San Luis Potosí, Shenyang (Tiexi)	No
BMW 2 Series	5	Chennai, Leipzig, Rayong, San Luis Potosí, Shenyang (Tiexi)	No
BMW X1	5	Araquari, Chennai, Rayong, Regensburg, Shenyang (Tiexi)	No
BMW X3	5	Chennai, Rayong, Rosslyn, Shenyang (Tiexi), Spartanburg	No
BMW X5	5	Araquari, Chennai, Rayong, Shenyang (Dadong), Spartanburg	No
BMW 5 Series	4	Chennai, Dingolfing, Rayong, Shenyang (Dadong)	No
BMW X7	3	Chennai, Rayong, Spartanburg	No
BMW i5	3	Dingolfing, Rayong, Shenyang (Dadong)	No
BMW iX1	3	Chennai, Regensburg, Shenyang (Tiexi)	No
BMW 4 Series	2	Dingolfing, Munich	No
BMW 7 Series	2	Chennai, Dingolfing	No
BMW X4	2	Araquari, Spartanburg	No
BMW X6	2	Rayong, Spartanburg	No
MINI Cooper	2	Oxford, Zhangjiagang (Spotlight)	No
MINI Countryman	2	Leipzig, Rayong	No
BMW 1 Series	1	Leipzig	Yes
BMW 8 Series	1	Dingolfing	Yes
BMW X2	1	Regensburg	Yes
BMW XM	1	Spartanburg	Yes
BMW i3	1	Shenyang (Tiexi)	Yes
BMW i4	1	Munich	Yes
BMW i7	1	Dingolfing	Yes
BMW iX	1	Dingolfing	Yes
BMW iX2	1	Regensburg	Yes
BMW iX3	1	Shenyang (Dadong)	Yes
MINI Aceman	1	Zhangjiagang (Spotlight)	Yes
MINI Cooper Convertible	1	Oxford	Yes

Appendix B — Calculation rules

Metric	Result	Rule
Comparable model lines	27	Count named model lines in the 2025 production programme after excluding “BMW M” and excluding Debrecen from the comparable matrix.
Comparable sites	14	Fourteen BMW/MINI passenger-vehicle sites remain after excluding Goodwood (Rolls-Royce) and Debrecen (source discrepancy).
Model-site links	63	Count all model-to-site entries across the 27 comparable model lines.
Single-site share	44.4%	12 single-site lines ÷ 27 model lines.
Average listed sites	2.33	63 model-site links ÷ 27 model lines.
H1 2026 production gap	110,784	1,267,511 production − 1,156,727 deliveries. (BMW Group, 2026h)
H1 2025 production gap	31,014	1,238,608 production − 1,207,594 deliveries. (BMW Group, 2026h)
Inventory increase	€2.801bn	€22.171bn at 30 Jun 2026 − €19.370bn at 31 Dec 2025. (BMW Group, 2026h)

Appendix C — Source register

ID	Source	Date	Publisher	Link / locator	Use in project
S1	BMW Group Report 2025	12 Mar 2026	BMW Group	Open source	2025 vehicle-plant programme and drivetrain portfolios
S2	The BMW Production Network of the Future	24 Mar 2025	BMW Group	Open source	Network strategy, flexible plants and regional balance
S3	How Data Scientists are Shaping the Future of Electromobility / Gen6 battery production	22 Oct 2025	BMW Group	Open source	Gen6 battery production footprint and local-for-local architecture
S4	BMW Group’s newest, most innovative production site officially opens in Debrecen	29 Sep 2025	BMW Group	Open source	Debrecen opening, iX3 launch timing and on-site Gen6 battery assembly
S5	Major milestone: First pre-series vehicles for new BMW i3 roll off the assembly line at BMW Group Plant Munich	4 Feb 2026	BMW Group	Open source	Munich pre-series i3 milestone and conversion
S6	Strong demand following early order opening: BMW Plant Munich launches BMW i3 series production with steep ramp-up curve	6 Aug 2026	BMW Group	Open source	Munich i3 series production and 2027 BEV-only plan
S7	New BMW 7 Series ramps up in Dingolfing: Progressive luxury meets Neue Klasse technologies	1 Jul 2026	BMW Group	Open source	Neue Klasse technologies introduced into established 7 Series production
S8	BMW Group Plant Regensburg marks 40th anniversary with record production	2 Feb 2026	BMW Group	Open source	356,901 vehicles in 2025 and flexible multi-drivetrain production

ID	Source	Date	Publisher	Link / locator	Use in project
S9	Think global - act local for local: Where the BMW Group will make its next generation of high-voltage batteries	30 Jul 2024	BMW Group	Open source	Five Gen6 battery assembly locations and local-for-local principle
S10	BMW Group Half-Year Report to 30 June 2026	30 Jul 2026	BMW Group	Open source	H1 2026 production, deliveries, inventory, working capital and FCF
S11	Statement by Walter Mertl, Finance, conference call	30 Jul 2026	BMW Group	Open source	Management explanation of H1/Q2 working-capital effect and production exceeding sales
S12	BMW Group completes $1.7 billion U.S. investment and premieres all-new BMW X5	30 Jun 2026	BMW Group	Open source	Spartanburg expansion and Woodruff battery network
S13	First glimpse into production: BMW Group Plant Woodruff assembles high-voltage batteries for the new BMW iX5	27 Jul 2026	BMW Group	Open source	Woodruff Gen6 battery production timing and local coupling to Spartanburg
S14	The start of a new era. The new BMW iX3	5 Sep 2025	BMW Group	Open source	Neue Klasse iX3, Debrecen series production and China rollout
S15	Construction of high-voltage battery assembly gets underway at BMW Group Plant San Luis Potosí	3 May 2024	BMW Group	Open source	2027 Neue Klasse production and local battery assembly
S16	Erneuter Produktionsrekord für das BMW Group Werk Leipzig: 2025 liefen am Standort über 259.000 Fahrzeuge vom Band	7 Jan 2026	BMW Group	Open source	Exact 2025 Leipzig production total (259,430) and plant-level model/drivetrain evidence

Source classification: figures marked as reported are taken from BMW Group publications. Calculated figures are derived from those inputs. Analytical classifications and scenario-free structural tests were developed for this project.

Appendix D — Publication, independence and AI use

1. Independence

- This is independent research based on publicly available BMW Group disclosures and other cited sources.
- BMW figures are reproduced or transformed only as needed to support the analysis. Project charts, calculations and analytical classifications are the author’s own work.
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- Publication date: 28 August 2026.
- Core evidence cut-off: 10 August 2026.

3. Use of tools and generative AI

- Microsoft Excel was used for data preparation, calculations, model construction and visual analysis.
- Generative AI tools were used for limited research organisation, language editing and formatting assistance.
- Source verification, dataset construction, calculations and final analytical conclusions were reviewed by the author.